

Name: _____

Homework #06: Special Purpose Registers

Give all answers in **hexadecimal**. Refer to the code provided for this homework for all answers.

1. At what address in ROM is the instruction on LINE 46 (`MOV R0, #0`) stored at

Answer: _____

2. What is the machine code for the following instruction? `BX LR`

Answer: _____

3. What is the machine code for the following instruction? Line 145: `UDIV R0, R0, R5`

Answer: _____

4. What is the value (address) stored in the LR after line 59 (`BL GetSum`) is executed?

Answer: _____

5. What is the value (address) stored in the PC after line 99 (`BX LR`) is executed?

Answer: _____

6. What is the value (address) stored in the SP before line 54 (`PUSH {R5, R6}`) is executed?

Answer: _____

7. What is the value (address) stored in the SP after line 54 (`PUSH {R5, R6}`) is executed?

Answer: _____

8. What is the value (address) stored in the SP after line 74 (`POP {R5, R6}`) is executed?

Answer: _____

9. What value (address) stored in the LR lets you know that no subroutines have been called?

Answer: _____

10. What value (address) stored in the SP lets you know that the stack is empty?

Answer: _____